Matthew Woody

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EDUCATION

Sonoma State University — Expected May 2027
B.S. Applied Mathematics | GPA: 3.41 / 4.00 — Rohnert Park, CA

- *Relevant Coursework:* Partial Differential Equations, Modern Physics, Numerical Analysis, Mathematical Modeling, Linear Algebra, Data Structures, Probability Theory

Santa Rosa Junior College — May 2025
A.S. Mathematics & Natural Sciences — Santa Rosa, CA

RESEARCH EXPERIENCE

Brookhaven National Laboratory — June 2026 – August 2026
Atmospheric Science Intern — Upton, NY
Advisor: Dr. Mindy Deng

- Built a quality-controlled, event-based dataset integrating NEXRAD Level II polarimetric radar with minute-resolution CoURAGE surface measurements for 18 cold front passages across heterogeneous terrain and land cover regimes
- Developed and validated a Python-based radar feature tracking workflow using polarimetric quality control, connected-component filtering, sensitivity-tested thresholds, lag-correlation propagation estimates, and DBSCAN clustering to isolate frontal precipitation
- Retrieved and composited polarimetric radar-derived drop size and rainfall properties across four radar transects to evaluate variability with terrain, season, and Chesapeake Bay breeze occurrence, through time-range diagrams and spatial profiles
- Identified recurring mountain region enhancement in cold front precipitation with longer duration, broader occurrence of high instantaneous rain rates, and ~2-4× greater accumulation than urban region precipitation during the analyzed spring and summer cases
- Evaluated methodological uncertainty and radar sampling limitations including beam geometry, retrieval assumptions, event-to-event variability, and sensitivity to feature selection thresholds
- Project Report: "*Tracking Precipitation in the CoURAGE Domain*" (PDF)

University of California, Davis — October 2025 – May 2026
Undergraduate Atmospheric Science Researcher
Advisor: Dr. Matt Igel

- Designed and automated a parameter sampling workflow in Octave for interacting DoNUT cloud circulations, using physically constrained cloud parameter ranges, duplicate-safe indexing, and systematic output labeling across separation distances from 0-12 km
- Built Python dataframes from model outputs and applied linear, quadratic, and random forest regression to quantify how DoNUT parameters affected extrema
- Validated regression trends by comparing controlled one-parameter-at-a-time simulation sequences against predicted shifts in model response features

Brookhaven National Laboratory — January 2025 – March 2025
Atmospheric Science Intern — Upton, NY
Advisor: Dr. Mindy Deng

- Applied K-means and DBSCAN clustering to XSACR radar observations from the TRACER campaign to isolate features of Houston's bay breeze

- Developed Python workflows to extract leading edge position and propagation speed, generate time series statistics and compare frontal velocity with overall bay breeze wind speed

PRESENTATIONS

BNL Environmental Science & Technologies Department Intern Seminar, Upton, NY, August 2026. “Tracking Cold Front Precipitation in the CoURAGE Domain”. Technical presentation to atmospheric science researchers.
BNL Summer Symposium, Upton, NY, August 2026. “Tracking Cold Front Precipitation in the CoURAGE Domain”. Selected research presentation for multidisciplinary audience of laboratory interns.
SSU Annual Math Festival, Rohnert Park, CA, April 2026. “Unsupervised Machine Learning Techniques to Identify Bay Breeze Statistics” (poster)

OTHER EXPERIENCE

SSU Learning and Academic Resource Center August 2026 – Present
Peer Tutor Rohnert Park, CA

- *Courses Supported:* Physics I & II, Differential Equations, Calculus, Reasoning and Proof, Mathematical Modeling, Probability Theory, Descriptive Astronomy, Finite Math (in-class tutor)

Santa Rosa Junior College January 2026 – May 2026
Finite Math Teaching Assistant Santa Rosa, CA

Brookhaven National Laboratory May 2025 – August 2025
Summer Policy Student Upton, NY

- Utilized prior internship familiarity to support program engagement and ensure a positive experience

City of Petaluma June 2024 – August 2024
Water Resources Intern Petaluma, CA

TECHNICAL SKILLS

Languages: Python, Octave, C++
Scientific Python: Xarray, Py-ART, scikit-learn, Matplotlib, NumPy, pandas
Tools: Jupyter Notebook, NetCDF, LaTeX, Excel